

SG-SST RAG — Complete Execution Graph

Complete LangGraph execution · Hybrid retrieval · Relevance routing · Memory · Fallback

ONLINE EXECUTION FLOW



GRAPH REFERENCE

NODE TYPES

- LLM / model processing
- Retrieval / evidence processing
- Conversation memory
- Validation / decision
- State / formatting
- Fallback / no grounded answer

GRAPH STRUCTURE

Entry:

business_profile

Sequential pipeline:

- business_profile
- retrieval_long_term_memory
- expand_query
- retrieve
- normalize_documents
- rerank
- expand_parent_documents
- record_retrieval_trace
- retrieval_relevance_grading

Conditional route:

Evidence → format_context
No evidence → fallback_answer

Common final path:

save_conversation_turn
→ store_long_term_memory
→ format_result → END

KEY STATE FIELDS

question	user input
business_context	company + memory
retrieval_query	expanded query
raw_results	hybrid retrieval output
documents	validated documents
context	normative context
messages	LLM messages
answer	generated response
result	LangChainRagResult

Traces:

query expansion · reranking
relevance grading · retrieval

DEFINED BUT NOT WIRED

sufficient_context_route()
SufficientContextGate
self_refine_trace
query_rewrite_trace
multi_query experimental state

These fields do not alter the execution graph shown here.

PROMPTS & LLM USAGE

Company profile

Identifies information about the company mentioned by the user.
Recognizes activity, CIIU, workers and risk class.

Query expansion

Enriches the user's question with useful SG-SST terminology.
Helps the search find documents even when different wording is used.

Evidence validation

Reviews the retrieved documents before they are used in the answer.
Keeps only information related to the user's actual question.

Answer generation

Builds the final response using only the validated evidence.
Company context is included when it helps personalize the answer.

Fallback explanation

Explains why the assistant could not provide a reliable answer.
It may identify missing evidence, an unrelated question or an error.
If generation fails, a safe predefined response is returned.

Conversation context

Adds useful information from the current and previous conversation.
Helps maintain continuity without requiring the user to repeat details.

LANGUAGE MODEL

DeepSeek
Supports understanding, validation and response generation.